Assignment-1

Writing Guidelines: `

1. Use clean A4 sheets for your assignments, ensuring legible handwriting.

2. Include the assignment number, course with section number, your name, and student ID on the first/cover page.

3. Number each page for easy reference.

4. Write your student ID on the top right corner of every assignment page.

5. Clearly label all your answers with the correct corresponding numbers.

1. The steam plot represents the ages of both males and females.

Leaf (male)	Stem	Leaf (female)
625826	0	389778
956240	1	240153
59561	2	62560
26169	3	40892
79407	4	18328

Key: 1|2 represents 12

a) Determine median for both male and female. [4]
b) What is the highest 25% cut-off point for males and What is the lowest 25% cut-off point for females? [4]
c) Draw frequency ogive curves for both. [3]
d) Determine which dataset is more stable. [4]

2. Suppose a coach trained 150 teenagers fast bowlers in a camp. The coach randomly selects 20 bowlers and records their highest speed. Highest speed of 12 bowlers are given below: 110,135, 126-R, 105,119,127-R, 116,152,121,132,119,118+R, 130,115,120, 108+R, 116, 127, 117+R, 108. [R = ***last digit of your student ID***]

a) Apply the best dispersion measure and comment on your findings. [5]
b) Estimate coefficient of skewness and interpret your finding [5]

3. Given below the age (in years) of 30 people a particular area,
18, 45, 29, 57, 34+R, 22+R, 48, 13, 60, 37, 23-R, 16, 32, 52, 28, 19, 25, 47, 39, 50, 41, 20, 30, 27, 58, 24, 35, 46-R, 69, 21

R = ***last digit of your student ID***

a) Construct a frequency distribution table. [4]
b) Calculate relative frequency, cumulative frequency, percentage frequency. [3]

4. The frequency distribution table of monthly income (in $000) of randomly selected individual are given below,

R = ***last digit of your student ID***

Income	10-20	20-30	30-40	40-50	50-60	60-70	70-80
Frequency	7	18	13	R	15	R	2

a) Determine the median of the dataset. [3]

b) Determine Q_1, $D_{7,}$ of the data set. [4]

c) Determine P_{90} of the dataset. [2]

d) Determine the average income of those selected individuals. [4]

e) Determine mean deviation. [5]